



Housing Price Over Time

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Our research goal: We set out to explore how InfluxDB could be used to analyze 26 years of U.S. housing price data and uncover what economic factors actually drive the market.

Why InfluxDB 3 Core?

A time-series database - it is purpose built for data with timestamps

Free and open source (MIT / Apache 2.0 license) which is the latest major version, released April 2025

Complete rewrite from earlier versions, built in Rust with Apache Arrow + DataFusion

Stores data as Apache Parquet which is a columnar format other tools (DuckDB, Spark) can read directly

SQL is the primary query language which is a big shift from version 2.x, which used a custom language called Flux

Supports high write throughput - designed for millions of data points per second

Native HTTP API write and query data without needing a client library

Tutorial Demo

Setup:

```
# Start server
.\influxdb3.exe serve
```

```
# Create token + database
.\influxdb3.exe create token --admin
.\influxdb3.exe create database housing
```

Ingest (Python):

```
Point("housing")
    .tag("series", "case_shiller")
    .field("value", float(val))
    .time(ts)
# 315 points written (2000–2026)
```

Query (SQL):

```
SELECT date_bin(INTERVAL '1 year', time)
    AS year, AVG(value)
FROM housing
WHERE series = 'mortgage_30yr'
GROUP BY year
```

Setup:

Start a local InfluxDB 3 server and create an authenticated database.

Ingest (Python):

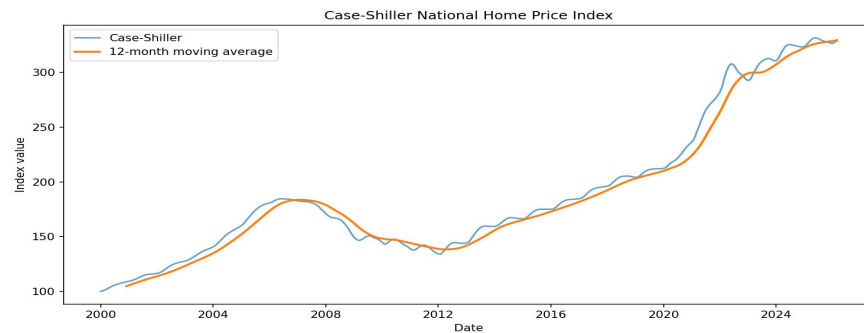
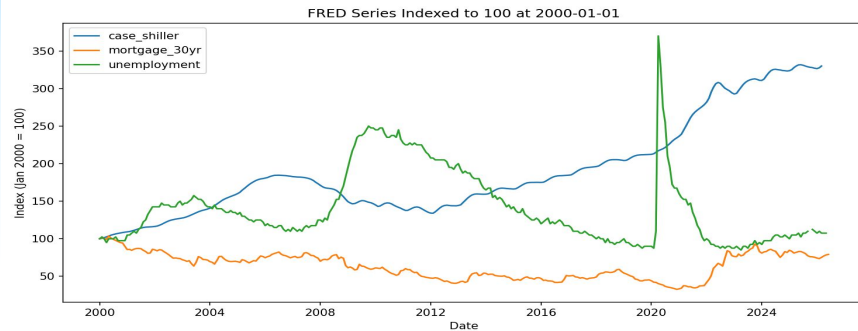
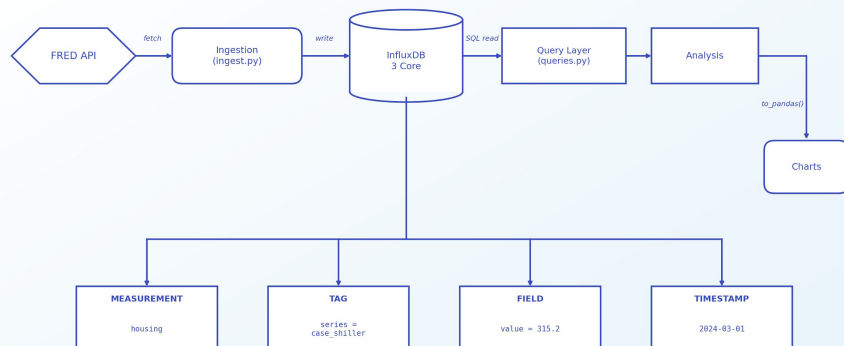
Each data point is written as a structured object with a tag identifying the series, a numeric field value, and a nanosecond-precision timestamp.

Query (SQL):

- InfluxDB 3 supports SQL with built-in time functions like `date_bin` for bucketing and downsampling time-series data by any interval.
- While InfluxDB's original native query language InfluxQL is still supported for backwards compatibility, SQL is now the recommended interface and InfluxQL is being phased out in InfluxDB 3.

Methodology

- We pulled three economic series from the FRED API. The Case-Shiller national home price index, the 30-year mortgage rate, and the unemployment rate covering monthly data from 2000 to 2026.
- A Python script wrote this data into InfluxDB 3 Core, which we then queried with SQL and pulled into pandas for analysis. We visualized the results with matplotlib.



Findings

- Housing prices are up 230% since 2000, a home that cost \$100k then would run you \$330k today
- Prices moved broadly inverse to mortgage rates for most of the period
- The 2008 financial crisis hit housing directly. COVID in 2020 caused a massive but brief unemployment spike while prices barely paused and then accelerated
- The rate-price relationship broke down after 2022. Mortgage rates spiked sharply, but home prices kept climbing instead of falling

Use cases

Capital One

- Serves 100M+ customers, generating massive streams of real-time data every second
- Uses InfluxDB to monitor apps, servers, and business metrics for fraud detection and system health
- Speed is critical: every minute of downtime directly impacts customer transactions and revenue
- InfluxDB's columnar, time-ordered structure lets engineers diagnose issues in seconds, not minutes



Tesla

- Manufacturing efficiency and minimizing downtime are top priorities
- Tags each data point by sensor type, machine ID, temperature, power consumption, etc.
- Engineers spot warning patterns early and schedule maintenance *before* machines break down
- Result: lower costs, less unplanned downtime, production lines that keep running



Strength/Weaknesses

Strengths

- Purpose-built for time-series data: columnar storage makes time-range queries significantly faster than relational or document databases
- Free and open-source (MIT + Apache 2 license): accessible for individuals, students, and enterprises alike
- Adopted standard SQL in v3, lowering the entry barrier for anyone already familiar with data querying
- Scales up to 1 million writes per second, which is suitable for real-world, high-volume applications
- Active community: 5,000+ companies, ~30,000 GitHub stars, Discord, Slack, and community templates

Weaknesses

- Single-purpose tool — only suited to time-series data, not general-purpose use cases
- Rearchitected repeatedly — query language went InfluxQL → Flux → SQL; Flux is now deprecated, leaving long-term users with obsolete skills and a painful migration path
- Clustering & multi-user auth missing from Core — production-grade collaboration and HA require upgrading to Enterprise

THANK YOU

Any Questions?

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